# HIMANSHU RAJ AMAN

**Senior Backend Engineer | PHP, Yii2, MySQL, AWS**

New Delhi, India (open to relocation: Germany / EU) | iamhimanshu7102@gmail.com | +91 6202328082 | linkedin.com/in/himanshurajaman | github.com/iamhimansu

## SUMMARY

Senior Backend Engineer with 4+ years building national-scale academic governance systems in PHP and Yii2, serving 2,000+ institutions and 100,000+ concurrent users. Focus on configurable rule engines, staged processing pipelines, and large-batch performance work. Recent work cut a core batch process from 10 hours to under 20 minutes.

## TECHNICAL SKILLS

- **Languages:** PHP (Expert), SQL, JavaScript, Node.js, C++
- **Frameworks:** Yii2, Laravel, Express.js, React.js, jQuery, Bootstrap
- **Data and infrastructure:** MySQL (query optimisation, JSON functions, keyset pagination), Redis, MongoDB, AWS (EC2, ELB, S3), Docker, Git, GitLab CI, GitHub Actions
- **Architecture:** REST APIs, rule engines (JSONLogic, DMN hit policies), staged pipelines, job queues, unit testing, SSE streaming, SaaS multi-tenancy, Model Context Protocol (MCP)
- **Currently learning:** Go (concurrency, backend services)

## PROFESSIONAL EXPERIENCE

**Senior Backend Engineer** *Aug 2025 - Present*

*Samarth eGov (Ministry of Education, Govt. of India initiative)* | *New Delhi, India*

- Cut grace-marks batch runtime from 10 hours to under 20 minutes by moving to array-based processing, bulk CASE/WHEN SQL updates, keyset pagination, and MySQL JSON aggregation.
- Rebuilt result processing as a staged pipeline where each stage receives a shared context object and owns one concern: course-count validation, component evaluation, final-result verification, then persistence, with the stages covered by unit tests.
- Modelled component pass/fail across combined filters (aggregate vs component-level, required-to-pass, attendance, disciplinary flags), with a simulation mode for dry-run verification before publishing.
- Designed a configurable, block-based grace-marks rule engine using JSONLogic conditions and DMN-style hit policies, replacing hardcoded per-ordinance logic with composable rules; modelled pipeline limits at 5,000,000-student scale.
- Eliminated recurring lock-wait timeouts by splitting one long transaction into per-batch transactions with exponential-backoff retry and idempotency flags.
- Built Flux, an in-house PHP/SSE library streaming long-running job output to the browser through a detached-worker architecture, bypassing Apache's 60-second request timeout.
- Led a small backend team through an ERP module overhaul, adding multi-tenancy and code-review standards; held 99.9% uptime across seasonal traffic spikes using AWS ELB, MySQL indexing, and Redis.

**Backend Engineer** *Oct 2023 - Aug 2025*

*Samarth eGov* | *New Delhi, India*

- Built the UG/PG admissions backend for Delhi University, handling 100,000+ requests during peak submission windows.
- Designed a standardised marks-entry data model adopted by 2,000+ colleges across multiple Indian states.
- Built a priority-based job queue for result processing with per-job status tracking and audit logging; diagnosed priority starvation that left large batches indefinitely deferred.
- Built parallel PDF report generation with chunked pipelines that preserve sort order; fixed mPDF backtrack-limit failures through incremental rendering.
- Exposed UIMS services to AI tooling through MCP servers in PHP, built on a reflection-based method index.

**Junior Backend Engineer** *May 2022 - Oct 2023*

*Samarth eGov* | *New Delhi, India*

- Integrated the National Academic Depository (NAD) API for secure academic record uploads and built template parsers for document automation.
- Built student feedback systems with automated Excel reporting using PHP Spreadsheet.

## PROJECTS

- **Process Manager** (PHP, Shell): multi-processing library using parent-child process coordination to queue and execute jobs over large datasets.

## EDUCATION

- **Master of Computer Applications**, Indira Gandhi National Open University, New Delhi (Jun 2023 - Present)
- **Bachelor of Computer Applications** (70%), Indira Gandhi National Open University, New Delhi (Jun 2019 - May 2024)